1.scenario: A user is required to enter a valid number in a form, but users sometimes input invalid data.

Logic:

1. Prompt the user to enter a number.
2. If the input is not a valid integer, display an error message.
3. Keep asking for input until the user enters a valid integer.
4. Once a valid integer is entered, proceed with further processing.

2.Scenario: A data analysis tool processes a list of numbers and needs to identify the most frequently occurring value.

Logic:

1. Read the list of numbers.
2. Create a dictionary or map to store the frequency of each number.
3. Traverse the list and update the frequency count for each number.
4. Identify the number with the highest frequency.
5. Return or display the most frequently occurring number.

3.Scenario: A text-processing application needs to compare words and check if they are anagrams (contain the same letters in a different order).

Logic:

1. Read the two input strings.
2. Remove any spaces and convert both strings to lowercase.
3. Sort the characters of both strings.
4. If the sorted versions of both strings are identical, they are anagrams

5. Otherwise, they are not anagrams.

4.Scenario: A speech analysis program needs to count the number of vowel sounds in a given input.

Logic:

1. Read the input string.
2. Define a set of vowel characters (a, e, i, o, u in lowercase and uppercase).
3. Initialize a counter to zero.
4. Loop through each character in the string:
    1. If the character is a vowel, increase the counter.
5. Return or display the total vowel count.

5.Scenario: A text-editing software includes a feature to reverse the order of words in a sentence for stylistic effects.

Logic:

1. Read the input sentence.
2. Split the sentence into a list of words.
3. Reverse the order of words in the list.
4. Join the words back together into a sentence.
5. Return or display the reversed sentence.

6.Scenario: A missing number is detected in a sequence of values stored in a database.

Logic:

1. Read the list of numbers.

2. Calculate the expected sum of numbers from 1 to n using the formula n * (n + 1) / 2.
3. Calculate the actual sum of numbers in the given list.
4. Subtract the actual sum from the expected sum to find the missing number.
5. Return or display the missing number.

7.Scenario: An ATM machine processes withdrawal requests and needs to ensure that users cannot withdraw more than their account balance.

Logic:

1. Read the user's account balance.
2. Prompt the user to enter a withdrawal amount.
3. Check if the withdrawal amount is less than or equal to the balance.
   If yes, deduct the amount and display a success message.
   If no, display an error message stating insufficient funds.

8.Scenario: A system needs to verify whether a given dataset contains duplicate entries.

Logic:

1. Read the list of values.
2. Create an empty set to store unique values.
3. Loop through each value in the list:
   1. If the value is already in the set, a duplicate exists.
   2. Otherwise, add the value to the set.
4. If duplicates are found, return True; otherwise, return False.

9.Scenario: A digital calculator includes a feature to sum the digits of a number for verification purposes.

Logic:

1. Read the input number.
2. Convert the number into individual digits.
3. Initialize a sum variable to zero.
4. Loop through each digit and add it to the sum variable.
5. Return or display the sum of the digits.

10.Scenario: A language-learning app wants to verify whether a given sentence is a pangram (contains every letter of the alphabet at least once).

Logic:

1. Read the input sentence.
2. Convert the sentence to lowercase and remove non-alphabetic characters.
3. Create a set to store unique letters.
4. Loop through each character in the sentence:
    1. If the character is a letter, add it to the set.
5. If the size of the set is 26 (i.e., contains all letters from 'a' to 'z'), the sentence is a pangram.
6. Otherwise, it is not a pangram.